

Chaitanya Mamatha Ananda

Compiler Optimization Engineer | C++ / LLVM | Ph.D. Candidate

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Summary

Computer Science Ph.D. candidate specializing in LLVM-based compiler optimization, with a research focus on improving memory footprint and locality. Experience in profile-guided, post-link binary, and AI-driven code-layout optimization.

Technical Skills

Programming: C++, C, Python, NumPy; Linux, Git

Compilers: LLVM, BOLT, Propeller, profile-guided optimization (PGO/FDO), post-link optimization, interprocedural optimization, basic block reordering and deduplication, code layout, heap locality

Performance: x86, Arm (AArch64), perf, hardware performance counters, profiling, benchmarking

ML for Compiler Optimization: AlphaEvolve, Vizier, MLGO, Evolution Strategies, heuristic search

Experience

Google | Student Researcher

Sep 2025 – May 2026

Compiler optimization team

Sunnyvale, CA

- Extended Google's **Propeller** with **AI-PROPELLER** for interprocedural code-layout optimization of warehouse-scale binaries. Used **AlphaEvolve** and hardware counters to guide optimization heuristic search.
- Reduced execution time by **1.6% on LLVM Clang** and **0.23% on a production Search workload**.

University of California, Riverside | Ph.D. Researcher

Sep 2022 – Present

Compiler optimization research; Advisor: Prof. Rajiv Gupta

Riverside, CA

- Developed **DeduBB** to deduplicate basic blocks across modules while supporting stack-manipulating code in **BOLT** and **Propeller** for x86 and AArch64. Reduced binary size by **up to 25.8%**, using profiles to target cold blocks and preserve performance.
- Developed **PreFix**, a profile-guided heap-layout optimization that groups hot objects to improve data locality. Reduced execution time by **21.7% on average**.

Indian Institute of Science | Research Assistant

Aug 2019 – May 2022

Parallel programming and scientific data analysis

Bengaluru, India

- Built a parallel programming model for PDE solvers using **Regent/Legion**. Developed a scientific-data anomaly detector using statistical and deep-learning methods.

Robert Bosch Engineering and Business Solutions | Intern

Spring 2021

Developed scripts to synchronize video and radar data collected during automated test driving.

Publications

DeduBB: Binary Code Size Reduction via Post-Link Basic Block Deduplication.

C. Mamatha Ananda, M. Afarin, R. Gupta, S. Tallam, H. Shen, and X. D. Li.

LCTES 2026 | [Paper](#)

PreFix: Optimizing the Performance of Heap-Intensive Applications.

C. Mamatha Ananda, R. Gupta, S. Tallam, H. Shen, and X. D. Li.

CGO 2025 | [Paper](#)

AI-PROPELLER: Warehouse-Scale Interprocedural Code Layout Optimization with AlphaEvolve.

C. Mamatha Ananda, R. Gupta, M. Trofin, A. Grossman, S. Tallam, X. D. Li, and A. Yazdanbakhsh.

MLArchSys 2026; CAIS AID-Wild 2026 (workshops) | [Paper \(arXiv\)](#)

Patent

Generate Better Code Layout Heuristics | Co-inventor, Google

Filed June 2026

Education

Ph.D. in Computer Science, University of California, Riverside

Sep 2022 – Present

B.E. in Computer Science and Engineering, Bangalore Institute of Technology, India

2021

Honors, Teaching & Service

Fellowship Dean's Distinguished Fellowship, UC Riverside

2022

Teaching Assistant Compiler Construction | Compiler Design

Fall 2023 | Summer 2024, 2025

Artifact Evaluation Committee Member PAW-ATM (SC26 workshop)

2026